board on which will the set will be used. I designed the pieces pictured for a board with 43 x 43mm squares, and chose a 28mm cylinder diameter which is 65% of 43mm. Because of the need to grip men to lift, move and place them, the cylinder diameter should decrease as a percentage of the board square's breadth the further that breadth is below 43mm. Similarly, that percentage can be increased as the board square's breadth increases above 43mm.

The diagram opposite shows the pieces at the exact size that I made them. If you want to turn a chess set with larger or smaller men than those dimensioned in the diagram included here, then you can magnify or reduce the piece designs you want from this article using a photocopier or by scanning and scaling in a computer. Alternatively, you could calculate the revised sizes from those specified using the appropriate multiplier.

Making the pieces other than the knights

Punching the centres for turning white-side men

Cutting the right-hand end of a workpiece flat with the long point of a skew. The workpiece is gripped in a scroll chuck's pin jaws

Scraping the recess in a white rook R1

Sawing the slot in a white bishop B1. The workpiece has already been finish-turned, polished and parted off

A 2mm wide parting-off tool

Chess men are best turned from dense woods. To differentiate the black and white sides you could use the same wood for both sides and apply paint or stain. I, however, prefer to use two woods of contrasting colours. Here though I've turned all the men from blackwood (*Acacia melanoxylon*), with the presence of the sapwood being the white side's side signature. I therefore carefully punched the ends of the white-side workpieces so that I could accurately centre them in the lathe for roughing.

In the previous four articles different chucking methods were shown. Although any of the four could be used to hold this article's cylinder-based men, I chose to use a pin chuck. I therefore roughed the individual workpieces to about 32mm diameter cylinders between centres.

For finish-turning I cantilevered each cylindrical workpiece from the pin jaws of a scroll chuck. After turning a workpiece to a cylinder of the designed diameter, I then cut the workpiece's right-hand end flat and far enough back to eliminate any centre point impression left from the roughing. These cuts are best performed with a skew's long point. For a skew's left-hand bevel to arc while remaining in a plane perpendicular to the lathe axis, right- handed turners should push the skew forward solely with the left hand. The finished length of the piece can then be marked from the right-hand end.

- The convex top of each king K1, the concave top of each queen Q1 or the recess in each rook R1 are turned as soon as the chucked workpiece has been brought to finished diameter, has had its right-hand end flattened and has been marked out.
- The slot in the bishop B1 can be marked out, sawn and chiselled by hand, or cut on a saw bench with a jig.

After finish-turning and sanding each cylinder-based man, I dechucked it without parting it off. I then rechucked them for polishing in the lathe as detailed in the first article. To part off after polishing I used a narrow (2mm wide) parting-off tool.

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